8TH GRADE MATH • SUPPORT NOTES

Unit 6: Functions

NC.8.F.1–5 • NC.8.EE.5, 8

Name: ____________________ Date: ____________________

Class: ____________________ Period: ____________________

★ The One Big Idea

A **function** is a rule where every input (x) gives **exactly one** output (y). Think of a machine: put a number in, one number comes out.

Words to Know

Input (x) — the number you put in

Output (y) — the number that comes out

Slope (m) — how steep the line is = rise ÷ run

y-intercept (b) — where the line crosses the y-axis (the start value)

Linear — makes a straight line

6.1 Introduction to Functions

NC.8.F.1

Vertical Line Test (VLT)

On a graph, if any **vertical line** hits the graph **more than once**, it is **NOT** a function.

f(x) Notation

f(x) just means “the output.” f(3) means “put 3 in for x.”

If $f(x) = 2x + 1$, then $f(3) = 2(3) + 1 =$ **7**.

You Try

$f(x) = 3x - 2$. Find $f(4) =$ ______

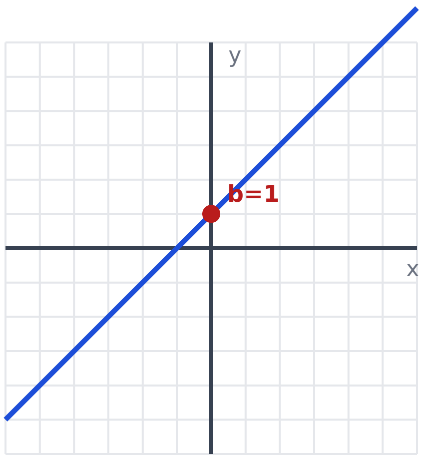
6.2 Slope & Linear Functions

NC.8.EE.5

★ y = mx + b

m = slope (steepness). **b** = y-intercept (where it starts).

$$\text{slope} = \text{rise} \div \text{run} = (\text{change in } y) \div (\text{change in } x)$$



y = x + 1 (slope 1)

Types of Slope

- **Positive**: goes up ↗
- **Negative**: goes down ↘
- **Zero**: flat line —
- **Undefined**: straight up |

You Try

From (0,2) to (3,8): slope = (8−2)÷(3−0) = ______

6.3 Tables → Equations

NC.8.F.4

Read a Table

1. **Slope:** how much y changes each time x goes up by 1.
2. **y-intercept:** the y value when x = 0.
3. Write y = (slope)x + (intercept).

I Do

x	0	1	2	3
y	2	5	8	11

y goes up 3 each step → slope = 3. At x=0, y=2 → b=2. So **y = 3x + 2**.

6.4 & 6.5 Real-World Meaning & Comparing

NC.8.F.2, NC.8.F.5

What Do m and b Mean in Real Life?

Slope (m) = the rate (per hour, per mile, each day).

y-intercept (b) = the starting amount (the fee before anything happens, the amount at time 0).

I Do

A taxi costs $y = 2x + 3$ (x = miles). The **3** is the start fee. The **2** is $2 per mile.

Compare Two Functions

Compare the **slopes** to see which grows faster. Compare the **y-intercepts** to see which starts higher.

6.6 Qualitative Graphs

NC.8.F.5

★ Reading a Story Graph

- Going **up** = increasing
- Going **down** = decreasing
- **Flat** = staying the same (constant)
- Steeper = changing faster

6.7 Systems of Equations

NC.8.EE.8

★ What Is a System?

Two equations together. The **solution** is the point (x, y) where the two lines **cross**.

Solve by Substitution

1. Get one equation as $y = \ldots$
2. Put that into the other equation.
3. Solve for x, then find y.

Example: $y = x + 1$ and $y = 2x - 1 \rightarrow x + 1 = 2x - 1 \rightarrow x = 2$, $y = 3$. Point **(2, 3)**.

Part 2 Practice

Functions

1. $f(x)=4x+1$, $f(2)$ = ______

2. Slope from (1,3) to (4,12) = ______

3. Table x:0,1,2 → y:5,8,11. Equation: y = ______________

4. In $y = 5x + 20$, the start value is ______ and the rate is ______.

5. Solve: $y = x + 2$ and $y = 3x$ → point ______________

★ ANSWER KEY ★

Guided Notes

6.1 You try: $f(4)=10$. 6.2 You try: slope=2. 6.3: $y=3x+2$.

Practice

1) 9   2) 3   3) $y=3x+5$ 4) start 20, rate 5   5) (1, 3)

End of Unit 6 support packet.